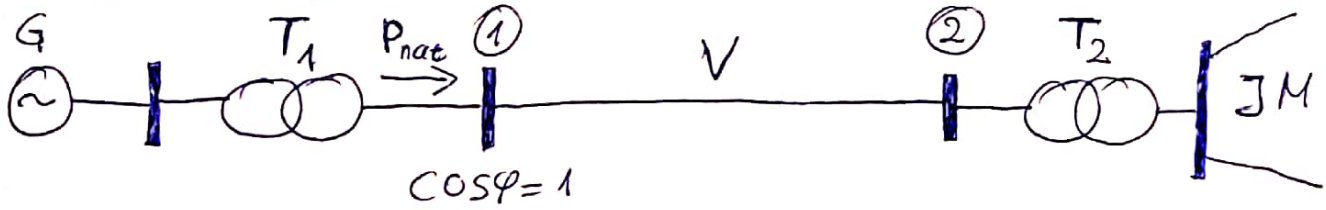
ЗАДАТАК ЗА ВЈЕЖБУ

13.05.2020.

Задатак 6:

а) На коју максималну удаљеност се за дати, једнотопло приказан трофазни систем, узевајући нацртану замјенску шему са слике, може пренијети природна снага вода номиналног напона 380[kV] са два проводника у сноп по фази карактеристичне импедансе $Z_c = 320[\Omega]$, да би систем остао стабилан на мале поремећаје?

b) На коју максималну удаљеност вода се може пренијети иста снага, да би систем остао стабилан на мале поремећаје, ако се на средини вода прекључи батерија кондензатор, чија реактанса компензује 50% реактансе вода 380[kV], дужине нађене под а)?



$S_{nG} = S_{nT1} = 500[MVA$

$U_{nG} = 15.75[kV]$

$m_{T1} = \frac{15.75}{400}\left[\frac{kV}{kV}\right]$

$x_{dG} = 160\%$

$x_{T1} = 15\%$

$U_1 = U_{nV} = 380[kV]$

$Z_c = 320[\Omega]$

$L_V = ?$

$S_{nT2} = 500[MVA]$

$m_{T2} = \frac{380}{220}\left[\frac{kV}{kV}\right]$

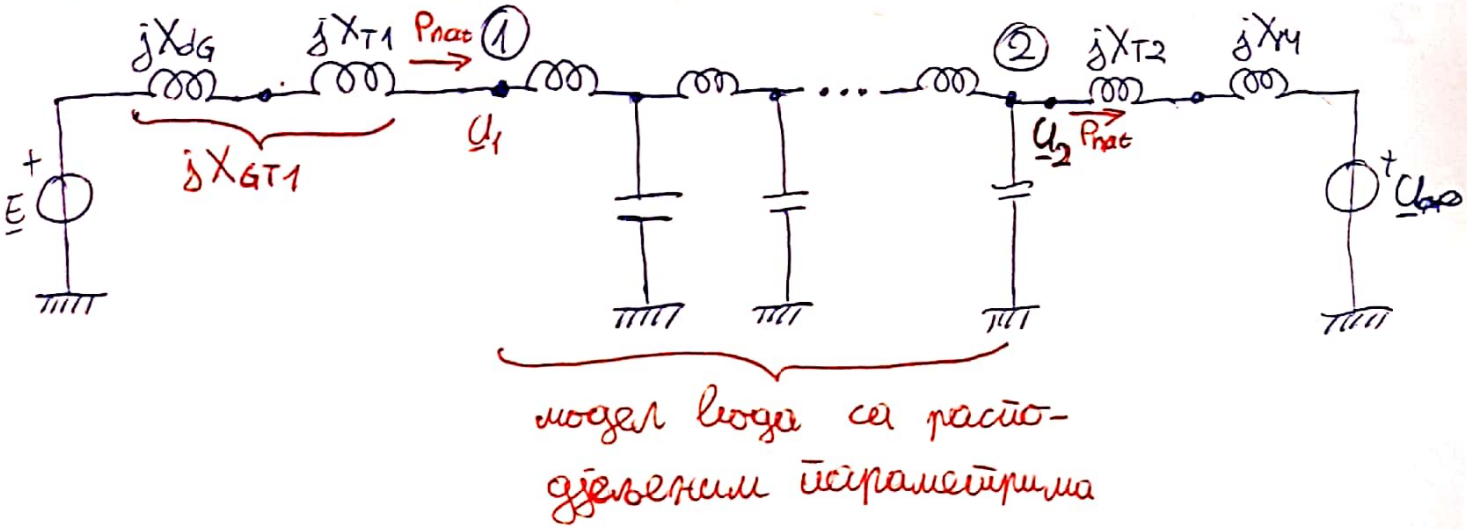
$x_{T2} = 10\%$

$U_b = const$

$f = const$

Scanned with Car

- Еквивалентна шема посматраног ЕЕС-а



а) Параметри елемената посматраног ЕЕС-а (сведени на ВН страну)

• генератор:

$$X_{dG} = \frac{x_{dG\%}}{100} \cdot \frac{U_{nG}^2}{S_{nG}} \cdot \frac{1}{m_{T1}^2} = \frac{160}{100} \cdot \frac{15.75^2}{500} \cdot \left(\frac{400}{15.75}\right)^2 = 512\,[\Omega]$$

• Тф-а:

$$X_{T1} = \frac{x_{T1\%}}{100} \cdot \frac{U_{nT1}^2}{S_{nT1}} = \frac{15}{100} \cdot \frac{400^2}{500} = 48\,[\Omega]$$

$$X_{T2} = \frac{x_{T2\%}}{100} \cdot \frac{U_{nT2}^2}{S_{nT2}} = \frac{10}{100} \cdot \frac{380^2}{500} = 28.88\,[\Omega]$$

$$X_{GT1} = X_{dG} + X_{T1} = 512 + 48 = 560\,[\Omega]$$

$$P_{nat} = \frac{U_{nV}^2}{Z_c} = \frac{380^2}{320} = 451.25\,[MW]$$

$$\underline{E} = \underline{U}_1 + \frac{\cancel{R_{GT1}}^{\,0} P_{nat} + X_{GT1} \cancel{Q}^{\,0}}{U_1} + j\,\frac{X_{GT} P_{nat} - \cancel{R_{GT}}^{\,0} Q}{U_1}$$

$$\underline{E} = \underline{U}_1 + j\,\frac{X_{GT} P_{nat}}{U_1} = 380 + j\,\frac{560 \cdot 451.25 \cdot 10^{\cancel{6}\,3}}{380 \cdot \cancel{10^3}}$$

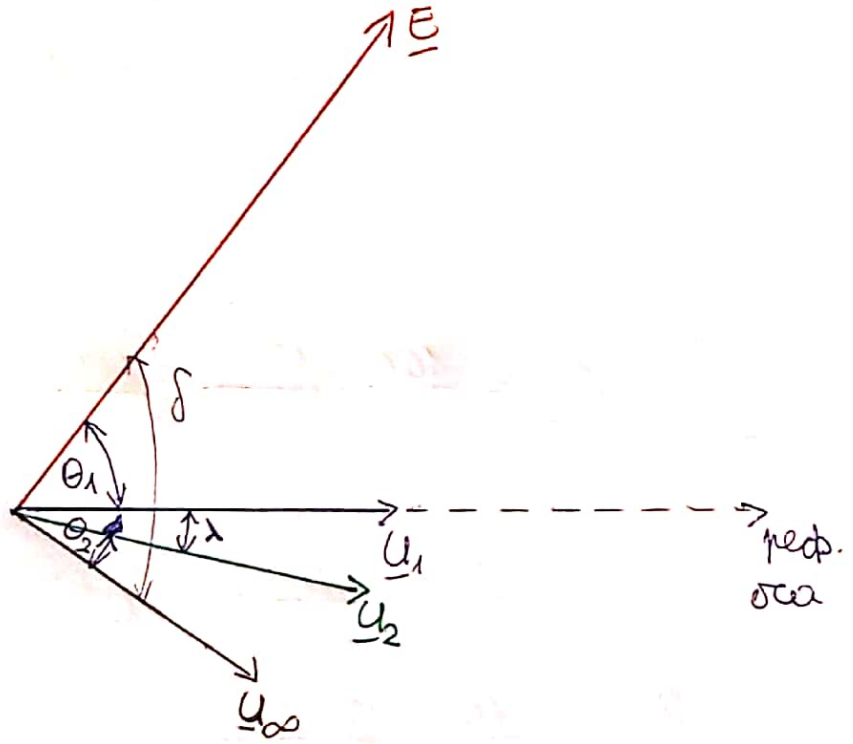
Scanned with Car

$$\underline{E} = (380 + j665)\,[kV] = 765.91 \angle 60.25^\circ\,[kV]$$

$$\underline{U}_\infty = \underline{U}_2 + \frac{\cancel{R_{T2}}^{\,0} P_{nat} + X_{T2} \cancel{Q}^{\,0}}{U_2} - j\,\frac{X_{T2} P_{nat} - \cancel{R_{T2}}^{\,0} Q}{U_2}$$

$$\underline{U}_\infty = \underline{U}_2 - j\,\frac{X_{T2} P_{nat}}{U_2} = 380 - j\,\frac{28.88 \cdot 451.25}{380}$$

$$\underline{U}_\infty = (380 - j34.295)\,[kV] = 381.54 \angle -5.16^\circ\,[kV]$$



- Пренесена дужина вода L_v одређује се из услова да је максимални фазни померај између фазора ЕМС $\underline{E}$ и напона на сабирницама бесконачне снаге $\underline{U}_\infty$ једнак 90°

$$\left.\begin{aligned} \Theta_1 + \lambda + \Theta_2 &= 90^\circ \\ \lambda &= 0.06\,L_v \end{aligned}\right\} \Rightarrow 60.25^\circ + 0.06\,L_v + |-5.16^\circ| = 90^\circ$$

$$L_v = \frac{90^\circ - 60.25^\circ - 5.16^\circ}{0.06}$$

$$\boxed{L_v = 409.83\,[km]}$$

Scanned with Car

b)

$$Z_c = \sqrt{\frac{l}{c}} \quad ; \quad c_0 = \frac{1}{\sqrt{lc}}$$ - брзина свјетлости

$$\Rightarrow \frac{c_0}{Z_c} = \frac{\frac{1}{\sqrt{lc}}}{\sqrt{\frac{l}{c}}} = \frac{1}{l} \Rightarrow l = \frac{Z_c}{c_0} = \frac{320}{300 \cdot 10^6} = \frac{320}{300} \cdot 10^{-6}$$

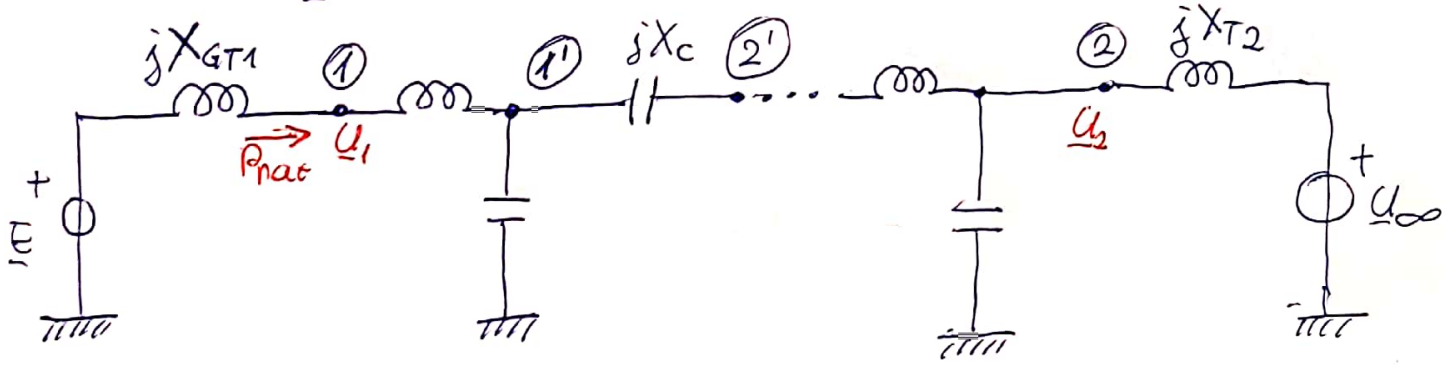
$$l = 1.0667 \cdot 10^{-3} \left[\frac{H}{km}\right]$$

$$L_V = 409.83 \, [km]$$

$$X_V = l \omega L_V = 1.0667 \cdot 10^{-3} \cdot 2\pi \cdot 50 \cdot 409.83 \Rightarrow X_V = 137.33 \, [\Omega]$$

- По услову задатка $X_c = 50\% \, X_V$

$$X_c = -\frac{1}{2} X_V = -\frac{1}{2} \cdot 137.33 \Rightarrow X_c = -68.67 \, [\Omega]$$



$P_1 = P_{nat}$
$Q_1 = 0$
$U_1 = U_{nv}$

Величине у тачки прикључења батерије кондензатора:

$P_{1'} = P_{nat}$

$Q_1' = 0$

$U_1' = U_1 \angle -\frac{\lambda}{2}$ (X_c је на средини вода)

- Услов статичке стабилности у овом случају је

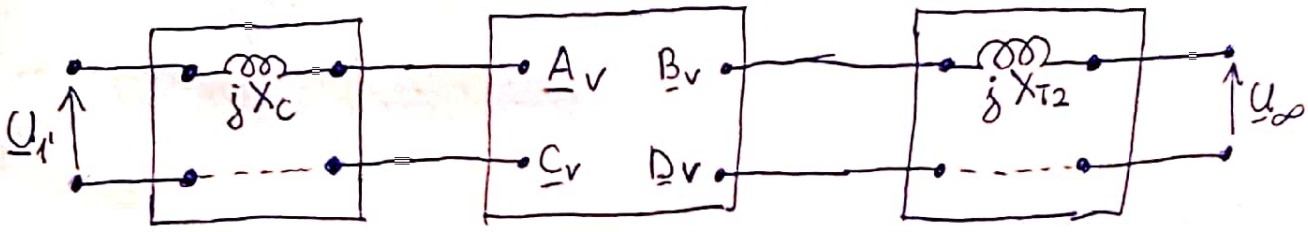
$$\Theta_1 + \frac{\lambda}{2} + \Theta_3 = 90^\circ$$

Scanned with Car

$\Theta_1 = 60.25^\circ \quad \measuredangle(\underline{E}, \underline{U}_1)$

$\Theta_3 \quad \measuredangle(\underline{U}_{1'}, \underline{U}_\infty)$

- Угао Θ_3 ће најлакше одредити користећи модел четворокраћника

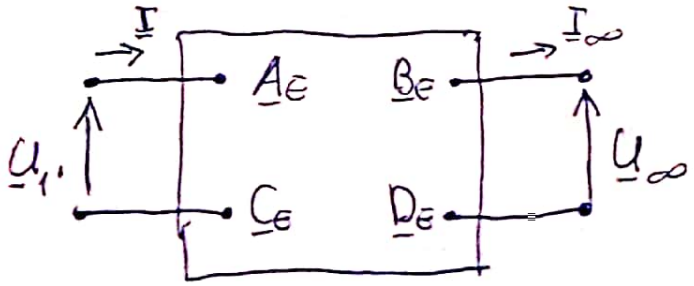


$$\begin{bmatrix} \underline{A}_E & \underline{B}_E \\ \underline{C}_E & \underline{D}_E \end{bmatrix} = \begin{bmatrix} 1 & jX_C \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos\left(\frac{\lambda}{2}\right) & jZ_c \sin\left(\frac{\lambda}{2}\right) \\ j\frac{1}{Z_c}\sin\left(\frac{\lambda}{2}\right) & \cos\left(\frac{\lambda}{2}\right) \end{bmatrix} \begin{bmatrix} 1 & jX_{T2} \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\left(\frac{\lambda}{2}\right) - \frac{X_c}{Z_c}\sin\left(\frac{\lambda}{2}\right) & j\left[Z_c \sin\left(\frac{\lambda}{2}\right) + X_c \cos\left(\frac{\lambda}{2}\right)\right] \\ j\frac{1}{Z_c}\sin\left(\frac{\lambda}{2}\right) & \cos\frac{\lambda}{2} \end{bmatrix} \begin{bmatrix} 1 & jX_{T2} \\ 0 & 1 \end{bmatrix}$$

$$\cancel{= \begin{bmatrix} \cos\left(\frac{\lambda}{2}\right) - \frac{X_c}{Z_c}\sin\left(\frac{\lambda}{2}\right) & j\left[X_{T2}\cos\left(\frac{\lambda}{2}\right) - X_{T2}\frac{X_c}{Z_c}\sin\left(\frac{\lambda}{2}\right) + Z_c\right.} \end{bmatrix}$$

$$= \begin{bmatrix} \cos\left(\frac{\lambda}{2}\right) - \frac{X_c}{Z_c}\sin\left(\frac{\lambda}{2}\right) & j\left[X_{T2}\cos\left(\frac{\lambda}{2}\right) - X_{T2}\frac{X_c}{Z_c}\sin\left(\frac{\lambda}{2}\right) + Z_c\sin\left(\frac{\lambda}{2}\right) + X_c\cos\left(\frac{\lambda}{2}\right)\right] \\ j\frac{1}{Z_c}\sin\left(\frac{\lambda}{2}\right) & \cos\left(\frac{\lambda}{2}\right) - \frac{X_{T2}}{Z_c}\sin\left(\frac{\lambda}{2}\right) \end{bmatrix}$$



Scanned with Cam

- Веза између напона и струја на почетку и на крају четвороокраја је

$$\begin{bmatrix} \underline{U}_{1'} \\ \underline{I}_1 \end{bmatrix} = \begin{bmatrix} \underline{A}_E & \underline{B}_E \\ \underline{C}_E & \underline{D}_E \end{bmatrix} \begin{bmatrix} \underline{U}_\infty \\ \underline{I}_\infty \end{bmatrix} \Rightarrow \begin{bmatrix} \underline{U}_\infty \\ \underline{I}_\infty \end{bmatrix} = \begin{bmatrix} \underline{A}_E & \underline{B}_E \\ \underline{C}_E & \underline{D}_E \end{bmatrix}^{-1} \begin{bmatrix} \underline{U}_{1'} \\ \underline{I}_1 \end{bmatrix}$$

$$\cancel{\begin{bmatrix} \underline{A}_E & \underline{B}_E \\ \underline{C}_E & \underline{D}_E \end{bmatrix} =} \quad \begin{bmatrix} \underline{U}_\infty \\ \underline{I}_\infty \end{bmatrix} = \begin{bmatrix} \underline{D}_E & -\underline{B}_E \\ -\underline{C}_E & \underline{A}_E \end{bmatrix} \begin{bmatrix} \underline{U}_{1'} \\ \underline{I}_1 \end{bmatrix} \quad \boxed{\underline{A}_E \underline{D}_E - \underline{B}_E \underline{C}_E = 1}$$

$$\underline{U}_\infty = \underline{D}_E \underline{U}_{1'} - \underline{B}_E \underline{I}_1$$

ако. $\underline{U}_{1'} = U_1 \angle 0^\circ \Rightarrow \underline{I}_0 = \dfrac{P_{nat}}{U_1} = \dfrac{451.25 \cdot 10^6}{380 \cdot 10^3}$

$$\underline{I} = 1187.5\ [A]\ \angle 0^\circ$$

$$\underline{U}_\infty = \left[\cos\left(\frac{\lambda}{2}\right) - \frac{X_{T2}}{Z_c}\sin\left(\frac{\lambda}{2}\right)\right] \underline{U}_{1'} -$$

$$-j\left[X_{T2}\cos\left(\frac{\lambda}{2}\right) - X_{T2}\frac{X_c}{Z_c}\sin\left(\frac{\lambda}{2}\right) + Z_c \sin\left(\frac{\lambda}{2}\right) + X_c \cos\left(\frac{\lambda}{2}\right)\right] \underline{I}$$

$$\cancel{\underline{U}_\infty = \left[380\cos\left(\frac{\lambda}{2}\right) - 34.295\sin\left(\frac{\lambda}{2}\right)\right] +}$$

$$\cancel{+j\left[47.25\cos\left(\frac{\lambda}{2}\right)\right.}$$

$$\underline{U}_\infty = \left[\cos\left(\frac{\lambda}{2}\right) - \frac{28,88}{320}\sin\left(\frac{\lambda}{2}\right)\right] \cdot 380$$

$$-j\left[28.88\cos\left(\frac{\lambda}{2}\right) - 28.88 \cdot \frac{(-68.67)}{320}\sin\left(\frac{\lambda}{2}\right) + 320\sin\left(\frac{\lambda}{2}\right) - 68.67\cos\left(\frac{\lambda}{2}\right)\right] \cdot 1.1875$$

Scanned with Car

$$\underline{U}_\infty = \left[380\cos\left(\frac{\lambda}{2}\right) - 34.295\sin\left(\frac{\lambda}{2}\right)\right] -$$

$$-j\left[-47.25\cos\left(\frac{\lambda}{2}\right) + 387.35\sin\left(\frac{\lambda}{2}\right)\right]$$

$$\underline{U}_\infty = \left[380\cos\left(\frac{\lambda}{2}\right) - 34.295\sin\left(\frac{\lambda}{2}\right)\right] +$$

$$+j\left[47.25\cos\left(\frac{\lambda}{2}\right) - 387.35\sin\left(\frac{\lambda}{2}\right)\right]$$

$$\underline{U}_\infty = U_\infty \angle -\Theta_3$$

$$\Theta_1 + \frac{\lambda}{2} - \Theta_3 = 90^\circ$$

$$\Theta_1 + \frac{\lambda}{2} - \operatorname{arctg}\frac{47.25\cos\left(\frac{\lambda}{2}\right) - 387.35\sin\left(\frac{\lambda}{2}\right)}{380\cos\left(\frac{\lambda}{2}\right) - 34.295\sin\left(\frac{\lambda}{2}\right)} = 90^\circ$$

$$\Theta_1 = 60.25^\circ$$

$$\lambda = 35.71^\circ \quad \text{- MATLAB}$$

$$L_V = \frac{\lambda}{0.06} = \frac{35.71}{0.06} \Rightarrow \boxed{L_V = 595.17\,[km]}$$

Scanned with Car